

Assignment 3

Build your own Language App



A way with words

Generate and explain idioms related to a keyword

Generate idioms

1. "Hold your horses" - Meaning to slow down or be patient.

Example: Hold your horses, I'm almost done with my presentation.

2. "Don't have a cow" - Meaning not to overreact to a situation.

Example: Don't have a cow, I'm sure the problem will be fixed soon.

3. "Cut the mustard" - Meaning to do something adequately or to meet the required standards.

Example: The team worked hard to cut the mustard and win the tournament.

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Introduction & motivation

"A Way with Words" is a simple language web app that leverages OpenAI's API to generate and explain idioms related to a given keyword. The primary objective of this app is to facilitate language learning and enhance understanding of idiomatic expressions.

Idioms play a significant role in language learning, as they contribute to the richness and depth of a language: they enrich communication, provide cultural insights, improve comprehension, expand vocabulary, enhance language proficiency, and encourage creativity in language use. Incorporating idioms into language learning programs can significantly contribute to a learner's overall success in mastering a new language.

For these reasons - and because learning idioms can be an enjoyable way to enrich one's vocabulary - I decided to go for the development of this app. The logo for the app features a literal depiction of the phrase *cat got your tongue*, which, as the reader is likely aware, refers to a situation where someone is at a loss of words or being unusually quiet. This representation highlights the potential benefits of incorporating visual learning experiences in the language acquisition process.

Overview of the app

The app works in a straightforward manner, requiring users to input a keyword for which they want to generate idioms. The app then sends a request to the ChatGPT API with the keyword, asking it to generate idioms and explanations related to the keyword. ChatGPT returns a list of related idioms along with their explanations, which are then displayed to the user.

"A Way with Words" is a simple adaptation of OpenAI's Quickstart GitHub repository. I followed the process shown in the document of A3 and gave it my own spin. I uploaded a zip file containing all of the code but the changes are contained in the files *generate.js*, *index.js* and *index.module.css*.

Generate.js

The main changes to this file were adding a parameter to fix the max number of tokens to 500 and the update of the function `generatePrompt` with the new prompt (figure below).

```
function generatePrompt(keyword) {  
  return `Provide a list of 3 English idioms related to the keyword 'sucess,' along with their meanings and examples of usage in sentences.  
  
  Keyword: Sucess  
  Idioms:  
  1. "The sky's the limit" - Meaning that there are no limits to what can be achieved, and success can be unlimited.  
  Example: With his talent and hard work, he knows that the sky's the limit for his career.  
  
  2. "Hit the jackpot" - Meaning to achieve great success, often unexpectedly or by luck.  
  Example: After years of hard work, the author finally hit the jackpot with their latest novel, which became a bestseller overnight.  
  
  3. "Rise to the occasion" - Meaning to perform well in a challenging situation or to meet high expectations, often leading to success.  
  Example: Despite being nervous, the athlete rose to the occasion and won the championship with a stunning performance.  
  
  Keyword: ${keyword}  
  Idioms:`  
}
```

Index.js

In this file I renamed everything related to the on screen display. The main challenge was managing to show the full output of chatgpt and make it visually appealing. The first issue was solved through the addition of two new classes (declared in the file `Index.module.css`) called *result_container*, which wraps the original *result* and ensures that it does not overflow and *example*, which consists of a splitted list of the lines of *result* (the split can be found in line 26 of this file).

```
<div className={styles.result_container}>  
  <div className={styles.result}>  
    {result.map((example, index) => (  
      <div key={index} className={styles.example}>  
        {example}  
      </div>  
    ))}  
  </div>  
</div>
```

Index.module.css

Minor changes were made in this file, all of them style related. Aside from the two new classes, I changed the size of some fonts, margins, headings and the icon of the app.

Collaboration with ChatGPT: Methodology and Benefits

In the completion of this assignment, I worked individually to achieve the objectives and meet the requirements. However, throughout the process, I relied on ChatGPT as a valuable partner and also got in touch with copilot.

ChatGPT assisted me in brainstorming and refining the original idea, adapting the code, and enhancing the overall quality of my work. This collaboration with ChatGPT allowed me to efficiently complete the assignment and saved me a lot of time. On a personal note, I got covid a few days after kingsday and I would have not even tried to do the assignment if it were not for the ease and the fun of working with this tool, since I had no time nor energy to face it.

Some Use-Case Examples



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1. "Go to the dogs" - Meaning to decline in quality or condition, often used to describe a decline in moral behavior.
Example: Ever since the new owners took over, the company has gone to the dogs.
2. "Let sleeping dogs lie" - Meaning to leave a situation alone and not cause any trouble or disruption.
Example: It would be best to let sleeping dogs lie and not bring up the argument from last week.
3. "Rain cats and dogs" - Meaning to rain heavily.
Example: It was raining cats and dogs, so we had to take shelter until the storm passed.



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Muscle

Generate idioms

1. "Put muscle into it" - Meaning to use strength or effort to do something.

Example: If you want to get the job done faster, you'll have to put some muscle into it.

2. "Gain muscle" - Meaning to make physical progress, usually through exercise.

Example: She has been gaining muscle in the gym and has seen amazing results in her physique.

3. "Show some muscle" - Meaning to demonstrate strength or power.

Example: The government will have to show some muscle if they want to get their message across.

Conclusion: Envisioning a More Refined Version

A more refined version of the "A Way with Words" app could incorporate additional features to enhance user experience and learning outcomes. One potential improvement would be to integrate the DALL-E 2 API, allowing the app to generate images alongside each idiomatic expression, such as the icon of the app. These images would serve as visual aids, creating a more engaging and memorable learning experience for users. I tried to do this but I did not manage to get anything worth showing in the given time frame. Nevertheless, it should not be hard.

Another possible enhancement would be the inclusion of interactive elements, such as a flashcard-based exercise where users match the correct meanings to given idiomatic expressions. This interactive component would encourage active learning and reinforce comprehension, ultimately promoting long-term retention of the acquired knowledge.

By incorporating these additional features or even the option for different game modes, the refined version of the app could offer an even more immersive and effective language learning experience, catering to different learning styles and preferences.